

## MATH4010 - ADVANCED PROBABILITY & STATISTICS

SPRING 2026

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### FINAL PROJECT SUBMISSION GUIDELINE

#### SUBMISSION AND DEADLINE

Each group must prepare **four items**:

1. One written report (PDF)
2. One program package (Python notebook, `.ipynb` / GitHub repository)
3. One presentation slide (PDF or PowerPoint)
4. One poster (PDF)

**Deadline:** all items (written report, code package, slides, and poster) must be submitted to Canvas **by 23:59 on Sunday, 31 May 2026**.

#### WRITTEN REPORT

- **Format.** The report must be in PDF format, with maximum **6 pages** (including references and contribution table). The report must be readable when printed (recommended: A4 size, standard margins, 11–12 pt font size).
- **Content.** The report must contain clear citations and references for any external sources used, including generative AI. Any graphs or figures must contain proper titles, labels, and descriptions.
- **Structure.** The report should follow the sample given in Appendix A. In particular, it must contain the **Contribution declaration** table.

#### PROGRAM PACKAGE

- **Format (choose one option).**
  - **Option A: Python notebook (`.ipynb`).**
    - \* Upload one copy to Canvas.
    - \* Upload one copy to Google Colab and insert the Colab link into the written report.
    - \* Ensure the link is accessible to instructors until your final grades are released.
  - **Option B: GitHub repository.**
    - \* Upload one copy (`.zip`) to Canvas.
    - \* Insert the repo link into the written report.
    - \* Ensure the link is accessible to instructors until your final grades are released.
- **Content.** The code must include all computations, experiments, and results reported in the written report.
- **Structure.** Organize the code into clearly labeled sections corresponding to each question.

#### PRESENTATION SLIDE

- **Format.** The slide must be in either PDF or PowerPoint format.
- **Content.** Focus on data collecting+processing, methodology, and experiment results.
- **Structure.** Divide the presentation into sections by question.

#### POSTER

- **Format.** The poster must be in 1 page A0, portrait, and PDF format.
- **Content.** Summarize the dataset, methodology, and key results.

## OTHER NOTES

It is strongly recommended to begin the project as early as possible. For any questions or clarifications, contact the instructor **by 23:59 on Wednesday, 27 May 2026**.

## UPDATE

Any changes will be posted here.

## APPENDIX A: QUESTION SHEET

### 1. Question 1.

Define two families of shapes parameterized by the dimension  $n$ :

- $A_n = \{\mathbf{x} \in \mathbb{R}^n \text{ such that } \|\mathbf{x} - \frac{1}{2}\mathbf{1}\|_2 \leq \frac{1}{2}\}$ , the largest ball inscribed in the unit box  $[0, 1]^n$ .
- $B_n = \{\mathbf{x} \in \mathbb{R}^n \text{ such that } \|\mathbf{x}\|_2 \leq 1\}$ , the unit ball.

In other word,  $A_n$  is the ball centered at  $(\frac{1}{2}, \dots, \frac{1}{2})$  and radius  $\frac{1}{2}$ .  $B_n$  is the ball centered at  $(0, \dots, 0)$  and radius 1.

#### 1.a.

- Visualize  $A_2$  and  $B_2$ .
- Compute the exact areas of  $A_2$  and  $B_2$  through direct calculation.
- Choose and implement algorithm(s) to estimate areas of  $A_2$  and  $B_2$ .
- Explain your algorithms and why you choose them. Report estimates, errors, and other metrics that you use.

#### 1.b.

- Visualize  $A_3$  and  $B_3$ .
- Compute the exact volumes of  $A_3$  and  $B_3$  through direct calculation.
- Use the same algorithm(s) from your part **1.a.** to estimate volumes of  $A_2$  and  $B_2$ .
- Report estimates, errors, and other metrics that you use. Compare against part **1.a.**

#### 1.c.

- Compute the exact volumes of  $A_n$  and  $B_n$  through direct calculation for any  $n \geq 1$ .
- Use the same algorithm(s) from your part **1.a.** to estimate volumes of  $A_{100}$  and  $B_{100}$ .
- Record any challenges and discuss on the performance of your selected algorithm(s) when  $N \rightarrow \infty$ . Suggest to revise / change to improve them if any.

### 2. Question 2

Let  $S$  be an action function given by  $S(x) = x^4$ . Let

$$f(x) = \frac{e^{-S(x)}}{Z}$$

be the associated probability density function.

- Estimate the normalization factor  $Z$  (by direct calculation or numerical algorithm).
- Choose at least three different symmetric proposal distributions and use Metropolis Algorithm to generate  $N = 1000$  samples following distribution  $f$ . Visualize.
- Explain your algorithm and report the numerical performance.
- Discuss on the performance across different symmetric proposal distributions, explain them and suggest any change to improve.

### 3. Question 3

Use a microphone (laptop, phone, or microcontroller with ADC) to record audio yourself. Use same setting (e.g., mono, 16-bit, at a fixed sampling rate (e.g. 16 kHz)).

**3.a. Data.** Record 10 clips of ambient noise (room noise, fan, traffic, etc) and 10 clips of music (songs played through a speaker, or you/someone playing or singing). Each clip the same length, at least 3 seconds long. Document the device and settings.

**3.b. Compute four numbers per clip.** Pick a bin width  $\Delta$  and use the *same*  $\Delta$  for all 20 clips. For each clip, compute:

- the discrete entropy  $\hat{H}_Q = -\sum_b \hat{p}_b \log_2 \hat{p}_b$ ;
- the differential entropy estimate  $\hat{H} = \hat{H}_Q + \log_2 \Delta$ ;
- the upper bound  $H_{\max} = \log_2 K$  where  $K$  is the number of bins;
- the Gaussian reference  $H_{\text{Gauss}} = \frac{1}{2} \log_2 (2\pi e \hat{\sigma}^2)$ .

**3.c. Discussion.** Report the mean and standard deviation of each quantity within each class. Compute the gap  $H_{\text{Gauss}} - \hat{H}$  for every clip and compare its typical value between the two classes. Which class is closer to Gaussian? Explain what this says using the maximum-entropy-given-variance result from lecture.

## APPENDIX B: WRITTEN REPORT SAMPLE

### MATH4010 Final Project: Group ##

#### Member list:

| No. | Name     | Student ID | Email        |
|-----|----------|------------|--------------|
| 1   | Jane Doe | V202310000 | Jane.d@email |
| 2   | John Doe | V202310001 | John.d@email |
| ... |          |            |              |

**Link to Google Colab:** [Insert link here]

#### 1. Question 1

Content.

#### 2. Question 1

Content.

#### 3. Question 1

Content.

#### 4. References

#### 5. Contribution declaration

| Task | Part          | Sub-task              | Tom | Rose | Kate | Bob |
|------|---------------|-----------------------|-----|------|------|-----|
| 1    | Write report  | Design questionnaires | ✓✓  | ✓    | ✓    |     |
| 1    | Write report  | Clean data            | ✓   |      | ✓    |     |
| ...  |               |                       |     |      |      |     |
| 1    | Write code    | Visualize histogram   |     |      |      | ✓   |
| 1    | Write code    | Convert data          |     |      |      | ✓   |
| ...  |               |                       |     |      |      |     |
| 1    | Prepare slide | Prepare slide         |     | ✓    | ✓    |     |
| 2    | ...           | ...                   | ... | ...  | ...  | ... |

*Note: Use ✓ to indicate contributors, use ✓✓ to indicate student in charge (who will answer Q&A about that sub-task). Each sub-task (row) selects at most 3 student, and each student is in charge at at least 1 sub-task.*

## APPENDIX C: GRADE RUBRIC OF WRITTEN REPORT AND PROGRAM PACKAGE

| Content                | Full | Level of Performance                                                                                                             |                                                                                                                                     |                                                                                                                                                                                                                                                  |
|------------------------|------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |      | Unsatisfactory                                                                                                                   | Satisfactory                                                                                                                        | Excellent                                                                                                                                                                                                                                        |
| (criteria)             |      | Missing required sections, unclear writing/structure, missing citations/references, or weak/incorrect interpretation of results. | Complete structure with clear writing and basic explanations; citations and figures mostly correct; minor issues may be acceptable. | Complete, well-organized, and clearly written; strong interpretation/discussion; figures and citations are thorough and professional. MUST include extra exploration, analysis and optimization instead of only a single accuracy or loss value. |
| <b>Written Report</b>  |      |                                                                                                                                  |                                                                                                                                     |                                                                                                                                                                                                                                                  |
| <b>Task 1</b>          | 30   | 0/5/10/15                                                                                                                        | 20/25                                                                                                                               | 30                                                                                                                                                                                                                                               |
| <b>Task 2</b>          | 30   | 0/5/10/15                                                                                                                        | 20/25                                                                                                                               | 30                                                                                                                                                                                                                                               |
| <b>Task 3</b>          | 30   | 0/5/10/15                                                                                                                        | 20/25                                                                                                                               | 30                                                                                                                                                                                                                                               |
| (criteria)             |      | Code is missing/incorrect, does not run end-to-end, or results are not reproducible/traceable to the report.                     | Runs and reproduces reported results; code is reasonably organized into sections with basic comments/outputs.                       | Clean, well-structured, and reproducible; clear sectioning and documentation; includes sanity checks and thoughtful extra exploration where appropriate.                                                                                         |
| <b>Program Package</b> |      |                                                                                                                                  |                                                                                                                                     |                                                                                                                                                                                                                                                  |
| <b>Code</b>            | 10   | 0/2/4/6                                                                                                                          | 8                                                                                                                                   | 10                                                                                                                                                                                                                                               |
| <b>All</b>             | 100  | ≤ 50                                                                                                                             | ≤ 83                                                                                                                                | ≤ 100                                                                                                                                                                                                                                            |

\* Note:

- The above rubric applies to the entire group. All group members will typically receive the same grade. However, individual member(s) who demonstrate significant or exceptional contributions may be considered for bonus points at the instructor's discretion. Bonus points, ranging from 1 to 5 (on a 100-point scale), are not guaranteed and may involve additional interview questions. They are awarded selectively and not necessarily granted to every group or every project.
- Any late/missing submissions without justification will result in 0 points for the entire group.
- Students who do not contribute to the project or are irresponsible for assigned tasks are subject to deductions, up to –100. This includes not coming to class/lab to work with group members, not being reachable during the project period, not accepting any task, or not completing tasks that appear in the attached contribution table under their name.

## APPENDIX D: GRADE RUBRIC OF PRESENTATION DAY

| Content                         | Full | Level of Performance                                                                                                               |                                                                                                                     |                                                                                                                                     |
|---------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                                 |      | Unsatisfactory                                                                                                                     | Satisfactory                                                                                                        | Excellent                                                                                                                           |
| Materials                       |      |                                                                                                                                    |                                                                                                                     |                                                                                                                                     |
| (criteria)                      |      | Missing required components; unclear/incorrect content; poor readability/organization; figures/tables are unlabeled or misleading. | Covers required content with mostly correct results and clear organization; visuals are readable with minor issues. | Polished and professional; clear and accurate narrative; strong visuals; insightful interpretation and takeaways.                   |
| Poster                          | 25   | 0/5/10/15                                                                                                                          | 18/20                                                                                                               | 25                                                                                                                                  |
| Slides                          | 25   | 0/5/10/15                                                                                                                          | 18/20                                                                                                               | 25                                                                                                                                  |
| Oral Presentation and Interview |      |                                                                                                                                    |                                                                                                                     |                                                                                                                                     |
| (criteria)                      |      | Unclear delivery; poor time management; cannot explain methods/results; major issues answering questions.                          | Mostly clear delivery and pacing; explains core ideas and results; answers most questions adequately.               | Confident, well-paced delivery; strong command of methods/results; engages audience; answers questions thoughtfully and accurately. |
| Oral presentation               | 25   | 0/5/10/15                                                                                                                          | 18/20                                                                                                               | 25                                                                                                                                  |
| Interview / Q&A                 | 25   | 0/5/10/15                                                                                                                          | 18/20                                                                                                               | 25                                                                                                                                  |
| All                             | 100  | ≤ 60                                                                                                                               | ≤ 80                                                                                                                | ≤ 100                                                                                                                               |

\* Note:

- The above rubric applies to the entire group. All group members will typically receive the same grade for poster/slides and the oral presentation.
- Interview/Q&A questions may be directed to individual members. If a member cannot answer questions about the project adequately, individual deductions may be applied at the instructor's discretion.